The State of Valuing Urban Ecosystem Services – An Informal Review

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November 8, 2026

1 Conceptualizing, Characterizing, and Measuring Urban Ecosystem Services

There is an extensive literature on the services that urban natural assets, such as parks, street trees, coastlines, wetlands, and small lakes and rivers, provide to people. Urban nature can provide heat control, improve air and water quality, mitigate flooding, offer a place to recreate, enhance physical and mental health, and promote social and cultural well-being (Keeler et al., 2019; Baró et al., 2016). The number of studies quantifying these services has been rising for two decades, and this trend accelerated after 2015 (Ma and Yang, 2025). Two recent reviews each cover more than 850 publications (Veerkamp et al., 2021; Ma and Yang, 2025). Cooling and recreation get the most attention. Local temperature regulation accounts for 36 percent of the studies, and recreation and aesthetic appreciation account for another 23 percent (Veerkamp et al., 2021). Large, publicly owned natural assets, such as parks, are studied most often (Veerkamp et al., 2021; Ma and Yang, 2025).

Despite the size and scope of the existing literature, there is no shared definition of "urban ecosystem service," and that ambiguity hinders researchers' ability to translate results into feasible actions (Tan et al., 2020). Half of the more than 300 papers they review do not define what an ecosystem service is (Tan et al., 2020). The ambiguity and variety of definitions may be due in part to the interdisciplinary nature of the topic, but definitions also vary within fields (Boyd and Banzhaf, 2007). In addition to the variety in 'what' is being measured, the methods used to measure ecosystem services vary tremendously (Veerkamp et al., 2021), with Ma and Yang (2025) identifying at least 47 methods for quantifying urban ecosystem services.

Flooding shows how far the words move. Haines-Young and Potschin (2010) lay out a "cascade": it starts with an ecological structure, which then provides some sort of function, and that function presumably serves and benefits people. Many urban studies follow this 'cascade' framework (Baró et al., 2016; Veerkamp et al., 2021). In Haines-Young and Potschin's (2010) example, a wetland is the ecosystem and it functions to slow surface water. Because flood control is beneficial to people, the wetland's function counts as an ecosystem service. In contrast, Gómez-Baggethun and Barton (2013) call the percolation of rain through soil and vegetation a function, and make the distinction that the flood control is the ecosystem service. Boyd and Banzhaf (2007) call the wetland itself the service, the flood control a benefit, and the ecosystem's functions as

inputs to services rather than services themselves. Even as the author of this paragraph, I get confused just writing it! The point is that across these three papers, preventing floods is a service in one, and a benefit in another, and a wetland is a structure in one and the service itself in another.[1]

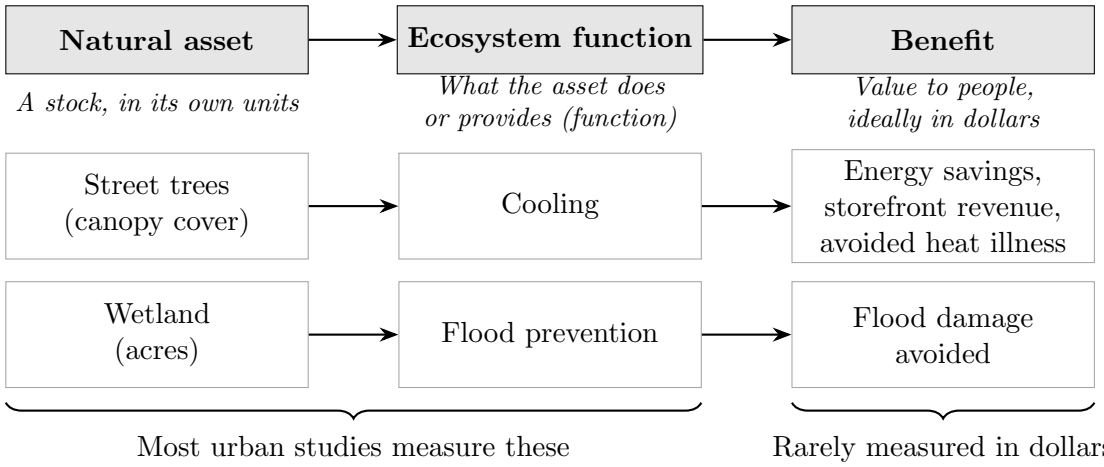
Additionally, the literature rarely measures the benefits of ecosystem services to people. This is another major hindrance to researchers' ability to influence decision-makers. Fifty-nine percent of reviewed papers quantify ecosystem properties, but a minority of current literature explicitly states the benefits of urban ecosystem services to people (Veerkamp et al., 2021). Even fewer papers measure benefits in monetary units, such as dollars. By not measuring the value of ecosystem services to people, researchers limit their ability to inform policy-makers, non-profits, businesses, and corporations about how to make informed decisions regarding urban natural assets. Research on urban nature-based solutions to climate change has underexamined their socioeconomic dimensions, and there is little city-scale empirical evidence on the economic benefits (McPhearson et al., 2025; Heris et al., 2021). The value of urban ecosystem services remains far more uncertain than what the services themselves are (Keeler et al., 2019).

Not understanding how ecosystem services benefit people is consequential for investing in new urban natural assets (*e.g.,* planting street trees), restoring degraded ones (*e.g.,* wetlands), and conserving current natural assets from development pressure. Without a dollar value on the benefit these natural assets provide, a city cannot run a benefit-cost analysis of planting street trees or restoring a stream, whether that analysis is to justify its own budget or to compete for grant money. It is also hard to know if businesses or corporations may have a private incentive to pay for the establishment or restoration of urban natural assets. Arguing that a city with a limited budget should allocate some of that budget to increasing ecosystem services requires knowing how the benefits of those services compare with other obligations of the city, such as schools, roads, and safety measures. Arguing that a business or corporation should invest in natural assets requires establishing how the ecosystem services from those assets will increase its profits, either through revenue gains or cost savings. Non-profits, such as land trusts and park conservancies, also restore and maintain urban natural assets. They do not need to make a profit, but they benefit when they can tell funders and city partners what a dollar spent on natural assets returns in terms of benefits to people.

The rest of this review proceeds as follows. First, I define natural assets, ecosystem functions, and benefits, and briefly address cultural and spiritual services (Section 2). Second, I summarize the well-established functions of urban green space, trees, and water, along with the dollar value of the benefits when available (Section 3). Third, I describe how unequally natural assets are distributed across urban neighborhoods (Section 4). Fourth, I point to a remaining gap in measuring the beneficial functions of urban animals (Section 5). Finally, I turn from measuring the functions of existing natural assets to asking who builds new ones and how research can inform that investment (Section 6).

[1]According to Google Scholar, Haines-Young and Potschin (2010) are cited 2,732 times, Gómez-Baggethun and Barton (2013) are cited 2,797 times, and Boyd and Banzhaf (2007) are cited 4,175 times. So, it's not as if any of these definitions are fringe!

Figure 1: Three Key Terms Regarding Ecosystem Services



Natural assets, ecosystem functions, and benefits as defined in this review. Most papers measure the asset and the function it provides (Veerkamp et al., 2021). Far fewer measure the benefit in dollars. This is only an illustrative diagram, and is not meant to be a census of the functions and benefits of street trees or wetlands!

2 What Is an Ecosystem Service?

The literature's inability to agree on a definition of "ecosystem service" limits its ability to build consensus on established results and on what remains to be understood. So, like so many before me, I will try to define three key terms related to ecosystem services so we can communicate with each other!

First, I define a natural asset as a stock of nature measured in its own units, such as acres of park, area of tree canopy, number of foxes, miles of trail, or surface area of a lake. A park, for example, is not an ecosystem service. It is a natural asset. The second is the ecosystem function that a natural asset provides, which I define as a function of a natural asset. A single natural asset may provide many functions. For example, a city park can function in many ways; it can create a cooler microclimate, a place to recreate, and control stormwater. The third key term is the function's benefit to people, which I define as the value of an ecosystem function to people. This benefit should, ideally, be measured in dollars so that it is easy to compare the benefits provided by natural assets (through their functions) with other benefits.

Figure 1 shows the three together. It is often possible to measure that a service is being provided without knowing its benefit. Consider a municipal government. It may know that trees (a natural asset) provide cooling (a function), but without some measure of the benefit of that cooling, it is hard to decide whether to spend the municipal budget on planting trees or on other beneficial services like roads or safety. Decision-makers who determine whether a city will have more or fewer natural assets therefore need a measure of the benefits of ecosystem services, so that managers can weigh those benefits against the other outcomes they are managing for.

Natural capital accounting puts heavy emphasis on a technical procedure for expressing monetary benefit in units of the natural asset (Fenichel et al., 2016; Fenichel, 2024). In this framework, one acre of park, one tree, or one mile of trail has a dollar value, in the same way that produced capital has a price. Fenichel and Abbott (2014),

for example, recover an accounting price for a pound of Gulf of Mexico reef fish. Eventually, having every natural asset priced this neatly would be amazing; natural capital accounting could be considered the gold standard for valuing a natural asset's benefits. However, it is technically very hard to connect a unit of benefit back to a unit of the asset. Doing so requires understanding the feedback loops between the natural asset, the human behavior that affects it, and the institutions that shape that behavior (Fenichel, 2024). Therefore, *measuring the benefit of urban natural assets perfectly should not be the enemy of measuring the benefit of these assets and their functions at all!*

A brief caveat: Cultural and spiritual ecosystem services fit less neatly into this framework. It is not always clear what the natural asset is, what function it provides, or how to put a dollar value on the benefit. A community may value a river for its spiritual significance, which is tied directly to the river, not to a service that flows from it. Species valued for their mere existence are less often valued formally because their non-market values are hard to derive (Maher et al., 2020), and Boyd and Banzhaf (2007) make a similar point about the emotional benefits of contact with nature. Culturally important species, which frequently matter to Indigenous Peoples and ethnic groups, are also more likely than other species to lack basic biological data (Reyes-García et al., 2023). Diep and McPhearson (2025) argue that the need to make a "business case" for nature-based solutions has given priority to economic valuation and undermined Indigenous Peoples' worldviews and other ways of relating to nature. However, I do not think that this should prevent us from valuing other ecosystem services in dollars. In my opinion, a dollar value is not meant to replace those other ways of valuing nature. It is instead meant to keep nature in the equation when a city weighs planting trees against schools and roads (*i.e.,* it is a useful unit, but not a more important one). I do not try to settle whether or how cultural and spiritual services should be valued in dollars. The rest of this review focuses on services whose assets, functions, and benefits are easier to define.

3 Established Functions of Urban Natural Assets

There are well-established ecosystem functions provided by urban green space, street trees, waterways, and wetlands (natural assets). These functions include urban heat mitigation, stormwater management, air purification, recreation, and health benefits (both physical and mental). Keeler et al. (2019) provide a summary table of urban natural assets and the ecosystem functions they provide that I find particularly useful for orienting to which asset does what function. I include their Table 1 as Figure 2 in this review.

Literature reviews also agree on which natural assets get the most attention. Green space of an unspecified type (30 percent) and parks and other natural green areas (26 percent) together make up more than half of the papers that Veerkamp et al. (2021) review. Abandoned land makes up only 2 percent, but may be the best opportunity for the establishment of new natural assets. Ma and Yang (2025) find a similar ordering. Woodlands, parks, and water bodies are studied more often than rain gardens, private gardens, and allotments, and green roofs and green walls are studied least.

The functions that are best understood are regulating functions. Microclimate, water, and gas regulation are the most frequently quantified (Ma and Yang, 2025),

Figure 2: Taken from Table 1 of Keeler et al. (2019).

Table 1 | Urban ecosystem services and nature-based solutions

Urban ecosystem services	Nature-based solutions						
	Street trees	Parks and open space	Engineered stormwater controls[a]	Green roofs	Waterways and wetlands	Coastal habitats[b]	Urban gardens
Urban air quality	X	X					
Carbon sequestration	X	X					
Coastal protection						X	
Urban heat and heat extremes mitigation	X	X		X			
Stormwater and wastewater management	X	X	X	X	X		
Urban water supply	X	X	X				
Riverine flood impact reduction		X	X		X		
Recreation opportunities		X			X	X	
Mental health	X	X					X
Urban agriculture		X					X

We reviewed literature pertaining to ten ecosystem services associated with seven nature-based solutions. Gaps in this matrix identify combinations where there was not sufficient literature to review the impact of a specific solution on a given service and/or the nature-based solution has been determined to be minimally effective with respect to a particular ecosystem service. [a]Such as bioswales, rain gardens, retention ponds. [b]Such as oyster reefs, mangroves, dunes, marshes.

What Keeler et al. (2019) call nature-based solutions, I call natural assets. What they call urban ecosystem services, I refer to as ecosystem functions.

and most studies measure them as physical changes, such as lower air temperatures, reduced stormwater runoff, or pollutants removed from the air and water (Veerkamp et al., 2021).

The benefits of these functions are much less understood. Only 3 percent of studies rely solely on what Veerkamp et al. (2021) call service indicators, which measure whether a natural asset meets people's demand for the function it provides. When studies do estimate a dollar value, simple methods dominate (Ma and Yang, 2025). Value transfer, which borrows values from earlier studies, accounts for about 47 percent of monetary valuations, and cost-based approaches, such as damage costs avoided or replacement costs, account for another 36 percent (Ma and Yang, 2025). Estimates also vary widely. Past studies report annual net benefits of $5 to $402 per urban tree, a range that suggests high uncertainty in the methods used to value trees, the returns to urban nature across cities, and which functions of a natural asset are included in a valuation study (Keeler et al., 2019). Of the ten services that Keeler et al. (2019) review, mental health has the most uncertain value, even though there is strong evidence that engaging with nature and living near urban green space improves public health (McPhearson et al., 2025). Finally, the value of urban nature is rarely compared with the built alternatives a city could pay for instead, such as air conditioning, seawalls, or levees (Keeler et al., 2019).

The following subsections elaborate on what we know about the functions of urban natural assets and the benefits they provide to cities and their residents.

3.1 Urban Heat Mitigation

The natural assets that regulate urban microclimates include trees and other vegetation, such as parks, green roofs, and green walls. A function these assets provide is cooling. Vegetation cools its surroundings through shade and evapotranspiration, which counteracts the urban heat island that makes cities hotter than the surrounding countryside

(Mohajerani et al., 2017; Wong et al., 2021). Greenery on the ground typically lowers peak surface temperatures by 2–9 °C (Wong et al., 2021), and in Phoenix, vegetation cools the surface by nearly 25 °C relative to bare soil on low-humidity summer days (Jenerette et al., 2011).

The benefit of cooling has been measured in dollars in several ways. Households in Phoenix have a marginal willingness to pay of $72 per month for a 1 °F reduction in outdoor temperature (Klaiber et al., 2017). Urban trees in major U.S. cities save energy for indoor cooling worth $539 million in 2016 (Heris et al., 2021). In my own work, I find that an increase in green space surrounding a business directly increases its revenue by half a percentage point on days at or above 100 °F (Creel, 2026).

Many more studies value trees through property prices (Sander et al., 2010; Netusil et al., 2010), but property prices capture a bundle of services that includes aesthetic beauty, as well. In Toronto, trees cool neighborhoods and save energy, but their total amenity value exceeds the value of these two services (Han et al., 2024). That study also measures value in units of the natural asset itself: a single tree adds 0.45 percent to property prices within a postal code.

3.2 Stormwater Regulation

For stormwater regulation, natural assets include trees and other permeable surfaces, such as parks and vegetated green infrastructure, that replace impervious pavement and roofs. The function is to intercept and absorb rainfall, which reduces the runoff that cities must otherwise manage (Veerkamp et al., 2021; Heris et al., 2021). Most of these studies model reductions in runoff or peak flow, often using the Storm Water Management Model (SWMM), and many examine vegetated features integrated into built infrastructure, such as rain gardens, swales, detention ponds, and constructed wetlands (Veerkamp et al., 2021).

A large majority of research on urban nature-based solutions focuses on water, especially flooding (McPhearson et al., 2025). Even so, it is still not clear which types of natural assets reduce flooding and where other solutions are needed (Diep and McPhearson, 2025). Trees, for example, intercept rainfall most effectively during small storms, so they may do little to mitigate flooding during extreme events (Heris et al., 2021).

As far as I am aware, there has been limited work valuing this service in dollars. Heris et al. (2021) estimate that urban trees in major U.S. cities intercepted about 2.6 billion cubic meters of rainfall in 2016, which they value at $425 million. That value is an avoided cost. In cities where stormwater and sewage share the same pipes, heavier stormwater flows raise treatment costs at the wastewater treatment plant and cause sewage to overflow into waterways. Heris et al. (2021) price each cubic meter of rainfall that trees intercept over impervious surfaces at the cost of controlling stormwater, about $2.58 (2019 USD), and they count only cities with these combined sewers, where the treatment plant is the beneficiary. The estimate therefore does not include avoided flood damage to homes and businesses.

3.3 Air Purification

For air purification, the natural assets are again trees and other urban vegetation. The function is to remove pollutants, such as particulate matter, from the air people breathe,

and tree species with larger leaf area or sticky or hairy leaves remove particulates at a higher rate (Keeler et al., 2019). Air quality regulation accounts for 13 percent of urban ecosystem service studies, and these studies most often measure function by tree characteristics or cover rather than by a specific type of green space (Veerkamp et al., 2021). In Lima, Peru, publicly managed urban trees remove about 27 tons of fine particulate matter (PM2.5) and 319 tons of coarser particulate matter (PM10) each year (Moreno et al., 2024). Where new trees are planted also matters. In a neighborhood of Perugia, Italy, converting marginal farmland into an urban forest would increase PM2.5 removal by 237 percent, while adding street trees would yield smaller gains but place trees closer to pollution sources, better aligning the function with where it is needed (Menconi et al., 2026).

At the scale of a whole city, however, the effect is small. Trees reduce city-wide air pollution by less than 1 percent to at most 5 percent, and reducing energy use or switching to cleaner energy may address particulate pollution more effectively (Keeler et al., 2019). Keeler et al. (2019) rate the potential value of urban nature for air quality as low and suggest shifting research away from goals such as mitigating air pollution with street trees and toward nature's contributions to recreation, social cohesion, and physical and mental health.

The benefit to people runs largely through health, which links this service to the health subsection below (Section 3.5). As far as I am aware, few studies value this function in dollars. Liu et al. (2020) come closest. In a choice experiment in Beijing, households would pay $2.04 per month (2017 USD) for one additional neighborhood park at the sample's average pollution level, but only $0.59 per month at the lowest pollution level. Willingness to pay for a new neighborhood park increases with residents' exposure to air pollution, whereas pollution has no significant effect on willingness to pay for a new city or national park. The authors interpret this as evidence that neighborhood parks directly purify the air for the communities around them (Liu et al., 2020). The willingness to pay is for the whole park, so it bundles air purification with recreation and everything else a park provides.

3.4 Recreation

For recreation, the natural assets are parks, trails, water bodies, and the tree-lined streets and sidewalks people walk along. The function is to provide a place to recreate and spend leisure time outside. Most outdoor recreation in the United States is local and centered on everyday activities, such as walking, playing with kids or pets, and socializing, so much of it likely occurs within cities (Creel and Fenichel, 2026). Which urban parks people use depends on what those parks offer. In the Twin Cities, parks with water features, more amenities, and trails, and parks in denser neighborhoods, get more visits (Donahue et al., 2018). In Beijing, large urban parks get the most visits in total, but neighborhood parks get more visits per unit of area (Zhang and Zhou, 2018). Demand for this function also rises when other leisure options close. Park visits were a critical source of leisure during the COVID-19 pandemic, when other ways to spend leisure time were forced closed (Geng et al., 2021; Venter et al., 2020; Day, 2020).

Recreation in parks is also social, and the societal trust built in city parks is another function of these natural assets that is not well understood at all. Visits to local parks are associated with more social ties among neighbors in inner-city Manchester

(Kaźmierczak, 2013), and urban parks in the Netherlands are places where ethnic groups mix and informal interactions can build social cohesion (Peters et al., 2010).

It is less clear how much of the visitation to parks is driven by the nature in a park rather than by its other features. Across New York City's 2,143 parks, geolocated social media posts, a proxy for visits, are more common in parks near public transit and bike routes and in parks with water bodies, athletic facilities, and impervious surfaces, but less common in parks with more green space (Hamstead et al., 2018). Hamstead et al. (2018) conclude that the kinds of green space in many parks may not motivate visits. Posts are also less common in parks in neighborhoods with a larger share of minority residents. The authors read this as a sign that parks in those neighborhoods may be less accessible or of lower quality, which I return to in the section on distribution below (Section 4).

The benefit of this service has been measured in dollars. In a working paper, Eli Fenichel and I use a travel-cost model to estimate the value of local outdoor recreation over seven three-year periods from 2003 to 2023, and find it is consistently between $5 and $6 per trip (2023 USD) (Creel and Fenichel, 2026). This value aggregates to about $93 billion per year nationally in 2021–2023. Park proximity and open space are also capitalized into property values. Reductions in crime "unlocked" almost $7 billion in property value near parks in Chicago, New York, and Philadelphia, while about $10 billion remains locked in (Albouy et al., 2020). Stated preference studies can value the quality of urban green space. Germans would, on average, pay €20.25 per month to make the green space closest to them more natural (Bronnmann et al., 2023).

Finally, Wu et al. (2023) estimate that about 97 percent of U.S. metropolitan areas have too little open space in their developed areas, so conserving more open space there would raise social welfare.

3.5 Health – Physical and Mental

For health, the natural assets are parks, trees, and green space more broadly. Health is better thought of as a benefit than a function, because several of the functions above deliver it (Sections 3.1, 3.3, and 3.4). Green space cools the places where people live, removes some pollutants from the air, and offers opportunities for physical activity, social contact, and rest from stress; each of these functions can improve physical and psychological health (Jennings and Gaither, 2015; Keeler et al., 2019). The benefit to people is fewer heat-related deaths and hospital visits, along with better mood and lower stress.

The evidence for physical health is strongest where it runs through cooling. Without vegetation, the urban heat island increases thermal stress on residents, which drives morbidity and mortality (Wong et al., 2021). Among elderly residents of eight Michigan cities, the odds of dying from cardiovascular disease during extreme heat rose by about 17 percent in ZIP codes that were 91 percent non-green space, but did not rise in ZIP codes that were 39 percent non-green space (Gronlund et al., 2015). Using emergency medical service records from Northern California, Bian (2025) finds that neighborhood-level extreme heat harms health, and policy simulations suggest that planting trees, especially in less vegetated areas, protects urban residents. The loss of trees has also been linked more directly to deaths. When the emerald ash borer killed 100 million trees across 15 U.S. states, the infestation was associated with an additional 15,080

cardiovascular deaths and 6,113 deaths from lower respiratory illness from 1990 to 2007 (Donovan et al., 2013).

The evidence for mental health is broad but less precise. There is strong evidence that engaging with nature and living near urban green space improves public health (McPhearson et al., 2025), and active use of green space is associated with better psychological well-being (Jennings and Gaither, 2015). Urbanization is associated with higher levels of anxiety and depression, so urban nature that improves mood and cognitive function is likely most valuable in cities where stress is high, and cross-sectional work suggests that nature experience may benefit people from disadvantaged backgrounds more than those from advantaged backgrounds (Keeler et al., 2019). Heat also harms mental health directly. On days above 25 °C (77 °F), calls to telephone counseling helplines in Germany increase by 3.7 percent (Janzen, 2025), suggesting that the cooling function of green space may matter for mental health as well.

Most studies of nature and health are observational and cross-sectional (Donovan et al., 2013), and it is hard to disentangle green space from its typical location in wealthier neighborhoods. For example, greener neighborhoods are associated with longer telomeres, genetic structures linked to cellular aging, but the association is no longer significant once air pollution and neighborhood deprivation are taken into account (Ogletree et al., 2023).

To my knowledge, the dollar value of these health benefits has been measured mainly through cooling. Across 97 U.S. cities, current tree cover avoids about 296 heat-related deaths each year, worth $1.0–$2.4 billion annually using the value of a statistical life (VSL), along with avoiding about 555 hospitalizations, 4,600 emergency department visits, and 52,000 outpatient visits (McDonald et al., 2020). Avoided deaths account for 76 to 84 percent of the total value in that study, and avoided illness and electricity savings make up the rest. The value of trees in preventing heat-related deaths has also declined as air conditioning has spread (McDonald et al., 2020). I am not aware of studies that quantify the mental health benefits of urban green space in monetary terms. This matters for the last section of this review (Section 6.2).

4 Natural Assets are Unequally Distributed

Knowing that a natural asset provides a service does not tell us who benefits from it, and natural assets are not evenly distributed across cities. Poor communities and communities of color have less tree canopy (Zhou and Kim, 2013; Sims, 2023), and historically redlined neighborhoods across the United States have less green space today (Nardone et al., 2021). Across 37 U.S. metropolitan areas, neighborhoods that were redlined in the 1930s have about 23 percent tree canopy cover today, while neighborhoods given the best grade have nearly twice as much (Locke et al., 2020), building on earlier work in Baltimore (Grove et al., 2018). Protected open space follows the same pattern. In New England, communities in the lowest income quartile have just 52 percent as much nearby protected land as those in the most affluent quartile, and the disparity holds within urban areas (Sims et al., 2022). Access is not always the issue, though. In six Illinois cities, racial and ethnic minorities have less tree canopy in their neighborhoods but no significant difference in access to parks (Zhou and Kim, 2013).

The same neighborhoods are hotter. Historically redlined neighborhoods experience

higher heat exposure today (Hoffman et al., 2020; Wilson, 2020; Conzelmann et al., 2022), and people of color are exposed to higher surface temperatures (Hsu et al., 2021; Renteria et al., 2022). Holding segregation constant, non-Hispanic Black residents are 52 percent more likely than non-Hispanic white residents to live where there is little tree canopy and mostly impervious surface, and that likelihood rises with segregation (Jesdale et al., 2013). In Phoenix, the availability of vegetation's cooling service was strongly and positively related to neighborhood income in 2000 (Jenerette et al., 2011). Poorer communities also lose more labor income during heat waves (Behrer et al., 2021). These disparities are large enough to shape urban ecology itself, including bird communities in Los Angeles (Schell et al., 2020; Wood et al., 2023).

These disparities matter for valuation, because the benefit of an ecosystem service depends on who receives it, and how economists measure benefits can change who appears to benefit. In my work on local outdoor recreation, pricing travel time with a replacement wage rather than one-third of each person's own wage compresses the richest quintile's per-trip value from about 10 times the poorest quintile's to only 1.5 times (Creel and Fenichel, 2026). Ignoring complementary public goods, such as safety near parks, can understate the value of parks to minority households (Albouy et al., 2020). Preferences also differ. A preference for connecting with people, rather than with nature, in parks may be more common among people of color than among white people (Elmendorf et al., 2005). Finally, the distribution of natural assets depends on where you look. The tree cover surrounding a storefront is nearly uncorrelated with its customers' income, so people experience a more equitable distribution of tree cover where they shop than where they live (Creel, 2026).

5 Urban Animals – The Next Frontier of Ecosystem Service Measurement?

The established literature focuses mostly on green space, water, and permeable surfaces (Section 3). Much less is known about the services that animals and insects provide in cities. In a review of 317 publications from 1990 to 2017, Schwarz et al. (2017) report 944 mentions of a relationship between biodiversity and an urban ecosystem service, but identify only two studies that consider how animal traits affect ecosystem function. The relationship between species diversity and pollination is among the few that may have been tested ten or more times!

Outside cities, economists have begun to consider how animals are a natural asset whose functions, such as pest control, provide value to people. Bats provide biological pest control. When a wildlife disease killed them off, farmers increased their insecticide use by 31.1 percent and infant mortality rose by 7.9 percent in the affected counties (Frank, 2024); migratory birds control pests on farms and in forests, and a 10 percent decline in bird biodiversity lowers crop revenues in the United States by 1.1 percent (Noack et al., 2025); and when wolves entered a county in Wisconsin, deer-vehicle collisions fell by 24 percent, avoiding more than $375,000 in losses per county per year, mostly because deer changed their behavior around wolves rather than because wolves killed them (Raynor et al., 2021). Comparable estimates for urban animals and insects are rare (and maybe even non-existent).

Urban carnivores, such as coyotes and foxes, may be able to keep rats and other

vermin under control and, in doing so, could prevent the spread of disease. Rodents thrive in cities in part because natural predators are scarce there, and cities spend millions of dollars on rodent control to reduce public health risks. Boyd and Banzhaf (2007) anticipate this kind of service, listing "predator populations hostile to disease transmission" among the components of nature that protect health. The closest evidence of this benefit in cities comes from vultures in India, which removed livestock carcasses and crowded out the dogs and rats that spread disease (Frank and Sudarshan, 2024). When vultures collapsed, human death rates rose by more than 4 percent, a loss of $69.4 billion per year. The increase was larger in urban areas, where death rates rose by 0.68 per 1,000 people, compared with 0.48 per 1,000 in the full sample (Frank and Sudarshan, 2024).

Animals also provide recreation. Wardle (2025) uses the 2017 recognition of the Cassia crossbill as its own species, found only in southern Idaho, as a natural experiment and finds that it raised birders' consumer surplus by at least $73,857 a year, or about $277 per visitor. The Cassia crossbill lives far from any city, but urban birders are passionate and plentiful!

The major challenge of including animals in the urban ecosystem service literature is measuring the natural asset itself. Urban land is well measured from space. A meta-analysis of 326 studies that used remotely sensed images to map urban land conversion finds that urban land grew by 58,000 km^2 worldwide from 1970 to 2000 (Seto et al., 2011), and forecasts built on satellite maps of urban extent show where cities will grow into habitat in key biodiversity hotspots by 2030 (Seto et al., 2012). Tree canopy is mapped nationally from satellite imagery (Housman et al., 2023), and green space along streets has been tracked with satellite data across 190 large cities (Falchetta and Hammad, 2025). Satellite data cannot provide a census of foxes, nor can it show how many rats those foxes suppress in New York (at least it cannot yet, and if it could there would be major concern about the surveillance of people). Counting urban animals takes data collected on the ground. In metro Atlanta, for example, researchers are currently monitoring remote game cameras at 60 sites to measure where medium and large vertebrates live (Kohl et al., 2024), and in Los Angeles, Wood et al. (2023) combine ground-based bird surveys with remotely sensed habitat data. Birds are the easier case, because birders count them. Birders submit more than 100 million sightings to eBird each year (Wardle, 2025). I do not know of any large-scale equivalent citizen-science project for other species. Without an population survey, it is hard to measure an animal's function, let alone its benefits.

6 From Measuring Ecosystem Services to Building Natural Assets

So far, I have focused on measuring the services that existing natural assets provide. In this final section, I want to pivot to how cities can "build" more natural assets, both to increase the benefits received from ecosystem functions and to widen the number and type of people who receive them. Therefore, this section now shifts the focus from measurement to who has the ability, the power, and the money to make new urban natural assets.

6.1 Who Makes New Urban Ecosystem Services

New urban natural assets come from restoration, conservation, or, in the case of animals, possibly reintroduction. Restoration often means turning neglected or abandoned land into functional green space. Who pays for this kind of project is the central question.

Usually it is the municipal government, often in partnership with a non-profit or with an organization like a business improvement district. City governments typically own the land that could become green space, such as sidewalks and parks, but their budgets are limited, and they face coordination frictions with the private actors who may have both an incentive and the ability to pay (Toxopeus and Polzin, 2021).

Non-profits have been strong partners for cities in raising money for new natural assets and caring for existing ones. Private park organizations, which are primarily non-profits, account for more than 40 percent of park spending in St. Louis and Detroit, and for about a quarter in Philadelphia, Boston, and Pittsburgh (The Trust for Public Land, 2025). Urban Resources Initiative in New Haven and the MillionTrees programs in New York City and Los Angeles are good examples of engaging communities in planting, maintaining, and caring for trees (Chen et al., 2024).[2] There is also evidence that the money non-profits raise improves environmental quality. More watershed groups, and more donations to and spending by those groups, improved water quality (Grant and Langpap, 2019).

Non-profits have done much of the heavy lifting to date, but two barriers stand in the way of unlocking more private financing: first, the coordination of private and public financiers, and second, understanding the private benefit of the functions of natural assets. Strategies for coordinating the two should identify who benefits privately — such as real estate developers, homeowners, and retail stores whose shoppers hate being too hot — to co-finance. Private investors need a financial return, so private investment in natural assets occurs only when an investment can be "monetized." Private investment is not a substitute for effective public policy that protects and invests in natural assets, but it is a complement in the way that it can be an additional revenue stream to pay for new parks, trees, swales, and other urban natural assets. However, both barriers come back to measurement. A city or non-profit will have an easier time coordinating a co-financing agreement with businesses or insurance companies when it can explain what the private benefits will be.

Insurers are another such actor that may experience large private benefits from investment in natural assets, because natural assets that lower disaster and health risk lower their claims. In Canada, sixteen property and casualty insurers fund Nature Force, a program led by Ducks Unlimited Canada that restores wetlands to reduce flood risk (Kousky and Burley, 2025), and in Quintana Roo, Mexico, an insurance mechanism on coral reefs, which halve expected annual flood damages (Beck et al., 2018), now supports conservation funding in its efforts to manage disaster risk for the coast (Schelske et al., 2021).

[2] I cannot overstate how much Colleen Murphy-Dunning's work with Urban Resources Initiative influenced my dissertation work. My first week of orientation at the Yale School of the Environment in 2019 had me weeding invasives to restore an abandoned lot to a functional park because of her. My interest in urban ecosystems stemmed from this influential early experience.

6.2 How Research Can Inform Investment

Research can inform investment in new natural assets by measuring the public benefits that justify public spending and the private benefits that can bring in co-financing. On the public side, benefit-cost analysis can win grant money or justify a municipal budget that invests in natural assets. The 2023 revisions to Circulars A-4 and A-94, which guide federal benefit-cost analysis, addressed how ecosystem services should be included. Those revisions have since been rescinded (Gillingham, 2025; Sarinsky and Schwartz, 2025), but are still great guidance for researchers wanting to consistently value ecosystem services. Additionally, a city deciding whether it will best take care of its residents by planting trees or repairing roads would benefit by knowing the benefit of the trees in dollars.

Research can also show when natural assets pay for themselves through public finances. The loss of bats, which provide free pest control to farmers, lowers county property tax revenue by an average of $1.2 million per year, or 3.8 percent (Nakhmurina et al., 2025), implying restoration could similarly increase county revenue collected from property tax.

On the private side, research can identify who benefits privately from a natural asset and by how much. In my job market paper, I find that extreme heat lowers storefront revenue and that surrounding tree cover protects it (Creel, 2026). In the southern United States, the avoided revenue losses can cover the cost of moving a storefront to 10 percent surrounding tree cover in about four years (although revenue is not profit). Other studies have shown extreme heat has a persistent negative effect on household retail consumption (Lee and Zheng, 2025) and lowers labor productivity (Dasgupta et al., 2024; LoPalo, 2023), and that trees lower the energy needed to cool buildings (Heris et al., 2021; Han et al., 2024). These are the kind of private benefits that a city or non-profit could point to when asking a business to co-finance nearby natural assets.

I think the least understood private incentive is health. Health insurance companies hold a tremendous amount of money, and Medicare and Medicaid are among the largest items in the federal budget. Causal evidence that investing in natural assets lowers health care costs could potentially unlock a huge pool of money for creating new natural assets. We have seen that insurers are willing to invest in natural assets when those assets lower risk (Kousky and Burley, 2025; Schelske et al., 2021). However, as the health section above shows (Section 3.5), most of the dollar value of urban trees and parks for health comes from avoided deaths, and the mental health benefit has not been valued in dollars at all. A health insurer pays for hospital visits and treatment, so benefits to an insurer are the parts we have measured least.

7 A.I. Disclosure

I used Consensus.app as a complement to Google Scholar when finding some relevant literature I didn't already have in my personal Zotero. I used Claude Opus 5.5 to help remember where all my papers were in my Zotero library, which said what again, and as a drafting aid. All citations and references were checked and verified. All final content and interpretations are my own. And, as much as I'd like to have a scape-goat, all mistakes are also my own.

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